

Bluestock Mutual Fund Analytics Capstone Project

Final Project Report

1. Executive Summary

Bluestock Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, industry trends, and portfolio risk using data analytics and business intelligence techniques. The project integrates data engineering, financial analytics, risk modeling, and dashboard development into a single end-to-end analytical solution.

The project processes multiple mutual fund datasets including fund master information, NAV history, investor transactions, SIP inflows, category inflows, portfolio holdings, benchmark indices, and industry AUM statistics. Data was cleaned, validated, transformed, and stored for further analysis using Python, SQL, and SQLite.

Comprehensive exploratory data analysis was performed to identify trends in NAV growth, SIP inflows, investor demographics, geographic investment patterns, and category-wise fund performance. Advanced financial analytics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional VaR (CVaR), Rolling Sharpe Ratio, and Herfindahl-Hirschman Index (HHI) were calculated to evaluate risk-adjusted performance and portfolio concentration.

An interactive Power BI dashboard was developed to provide decision-makers with insights into industry performance, fund analytics, investor behavior, benchmark comparison, and SIP trends. The project demonstrates practical application of data analytics, business intelligence, and financial risk management techniques within the mutual fund industry.

2. Project Objectives

The primary objective of this project is to build an end-to-end Mutual Fund Analytics Platform that provides actionable insights into mutual fund performance, investor behavior, industry growth, and portfolio risk. The project aims to transform raw financial data into meaningful business intelligence through data engineering, analytics, and visualization.

Specific objectives include:

- Building a robust ETL pipeline for data ingestion, cleaning, validation, and transformation.
- Analyzing mutual fund performance using CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Studying investor behavior through demographic, geographic, and transaction-level analysis.
- Evaluating industry trends such as SIP inflows, AUM growth, and folio expansion.
- Developing advanced risk analytics including VaR, CVaR, Rolling Sharpe Ratio, and HHI concentration analysis.
- Creating an interactive Power BI dashboard for decision support and business intelligence.
- Implementing a simple fund recommendation engine based on investor risk appetite and historical risk-adjusted performance.

3. Data Sources

The project utilizes multiple datasets representing different aspects of the mutual fund ecosystem.

Fund Master Dataset:

Contains scheme metadata including AMFI code, fund house, category, benchmark, expense ratio, risk category, and launch details.

NAV History Dataset:

Provides historical Net Asset Value (NAV) data used for return calculations, trend analysis, and risk metrics.

Investor Transactions Dataset:

Includes SIP, lumpsum, and redemption transactions along with investor demographics, location, and income information.

AUM Dataset:

Contains industry and AMC-level Assets Under Management used to analyze market growth and fund house dominance.

SIP Inflows Dataset:

Provides monthly SIP contribution trends and account statistics.

Category Inflow Dataset:

Captures category-wise fund inflows and outflows across different time periods.

Portfolio Holdings Dataset:

Contains portfolio allocation data including stock-level holdings, sectors, and weights.

Benchmark Indices Dataset:

Includes benchmark performance data such as NIFTY 50 and NIFTY 100 used for comparative analytics.

4. ETL Design and Data Pipeline

The ETL (Extract, Transform, Load) pipeline was designed to ensure high-quality and analysis-ready datasets. Raw data was collected from multiple mutual fund sources and stored in CSV format.

Extract:

Raw datasets including fund master data, NAV history, investor transactions, portfolio holdings, benchmark indices, SIP inflows, category inflows, and AUM statistics were imported using Python and Pandas.

Transform:

Data cleaning procedures included handling missing values, removing duplicate records, standardizing column names, validating data types, converting date fields, and checking for logical inconsistencies. Additional derived metrics such as daily returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR were calculated during the transformation stage.

Load:

The cleaned datasets were stored in structured CSV files and SQLite tables for efficient querying and downstream analytics. These datasets served as the foundation for exploratory analysis, risk analytics, and dashboard development.

The ETL pipeline improved data quality, consistency, and reproducibility while enabling efficient analytical workflows.

5. Database Design

A relational data model was designed to organize mutual fund, benchmark, investor, and portfolio information. The database structure enables efficient integration of fund performance, investor transactions, benchmark comparisons, and portfolio analytics.

Primary Tables:

- Fund Master – Contains scheme metadata, category information, benchmarks, and expense ratios.
- NAV History – Stores daily NAV values used for return calculations and trend analysis.

- Scheme Performance – Contains performance indicators such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Investor Transactions – Records SIP, lumpsum, and redemption activities along with demographic information.
- Portfolio Holdings – Stores stock-level portfolio allocation and sector exposure data.
- Benchmark Indices – Contains benchmark performance information including NIFTY 50 and NIFTY 100.

Relationships were established using AMFI Code and Date fields wherever applicable. This structure supports scalable analytics and dashboard reporting while maintaining data consistency.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, quality, and trends present within the mutual fund datasets. Multiple visualizations and statistical summaries were created to identify patterns related to fund performance, investor behavior, industry growth, and portfolio allocation.

Key EDA Findings:

- Mutual fund Assets Under Management (AUM) showed consistent growth over the analysis period, indicating increasing investor participation in the mutual fund industry.
- SIP inflows demonstrated a strong upward trend, reflecting growing preference for systematic investing among retail investors.
- Equity-oriented categories received the highest inflows compared to debt and hybrid categories.
- Investor transaction analysis revealed significant participation from Tier-1 and Tier-2 cities, with younger investors contributing a large proportion of SIP investments.
- Geographic analysis identified concentration of investments in major states such as Maharashtra, Karnataka, Delhi, Gujarat, and Tamil Nadu.
- Portfolio analysis showed varying levels of sector concentration across schemes, highlighting differences in diversification strategies.
- Benchmark analysis indicated strong correlation between fund NAV movements and major market indices such as NIFTY 50 and NIFTY 100.

The EDA phase provided valuable insights that guided subsequent performance analysis, risk modeling, and dashboard development.

Figure 1: AUM Growth by Fund House

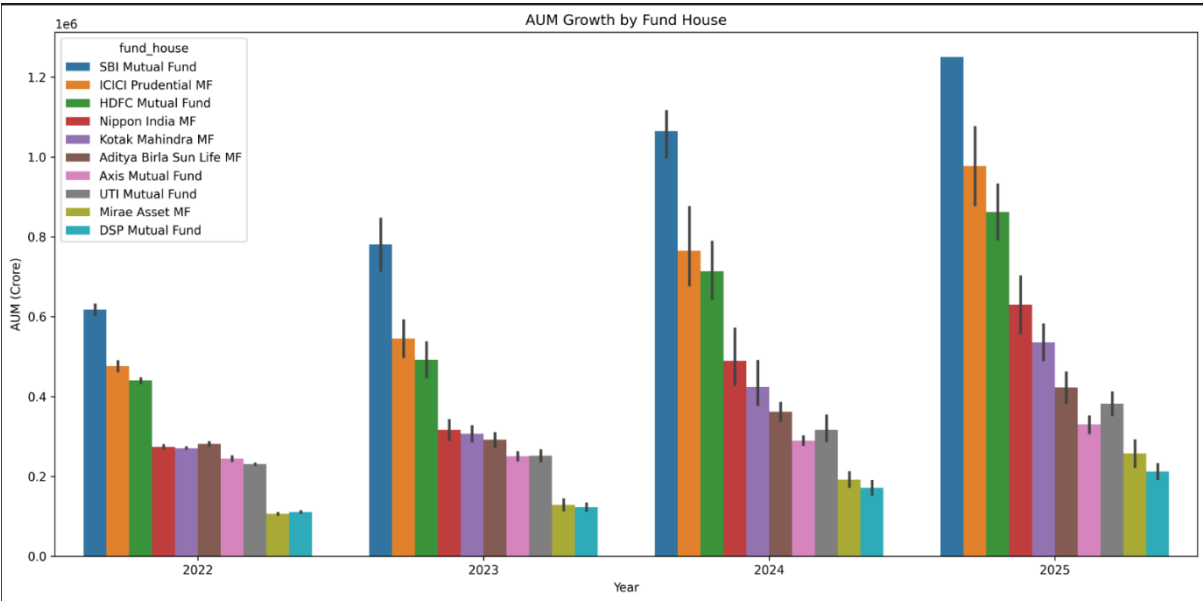


Figure 2: Monthly SIP Inflow Trend

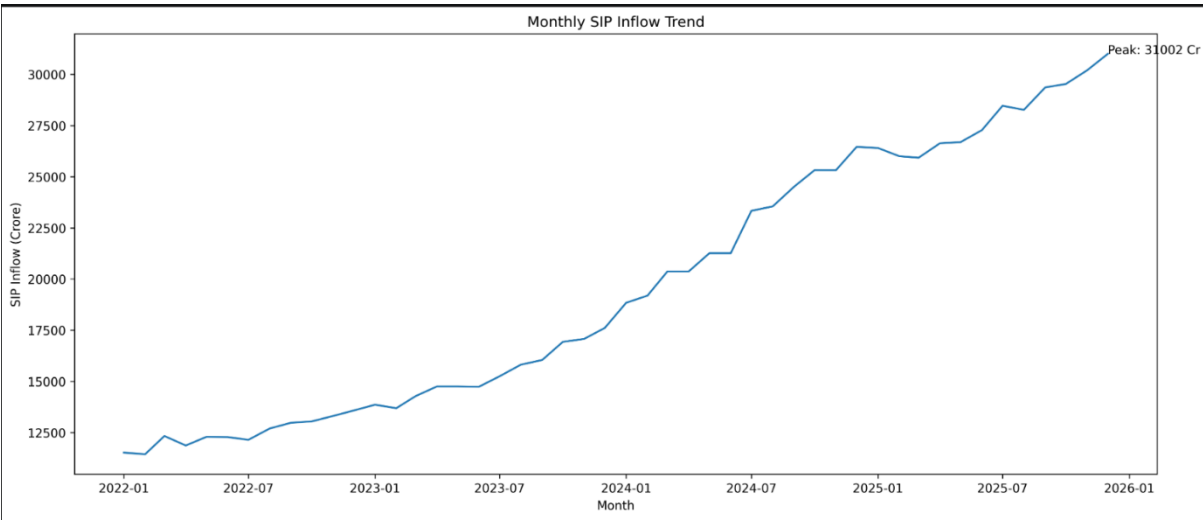
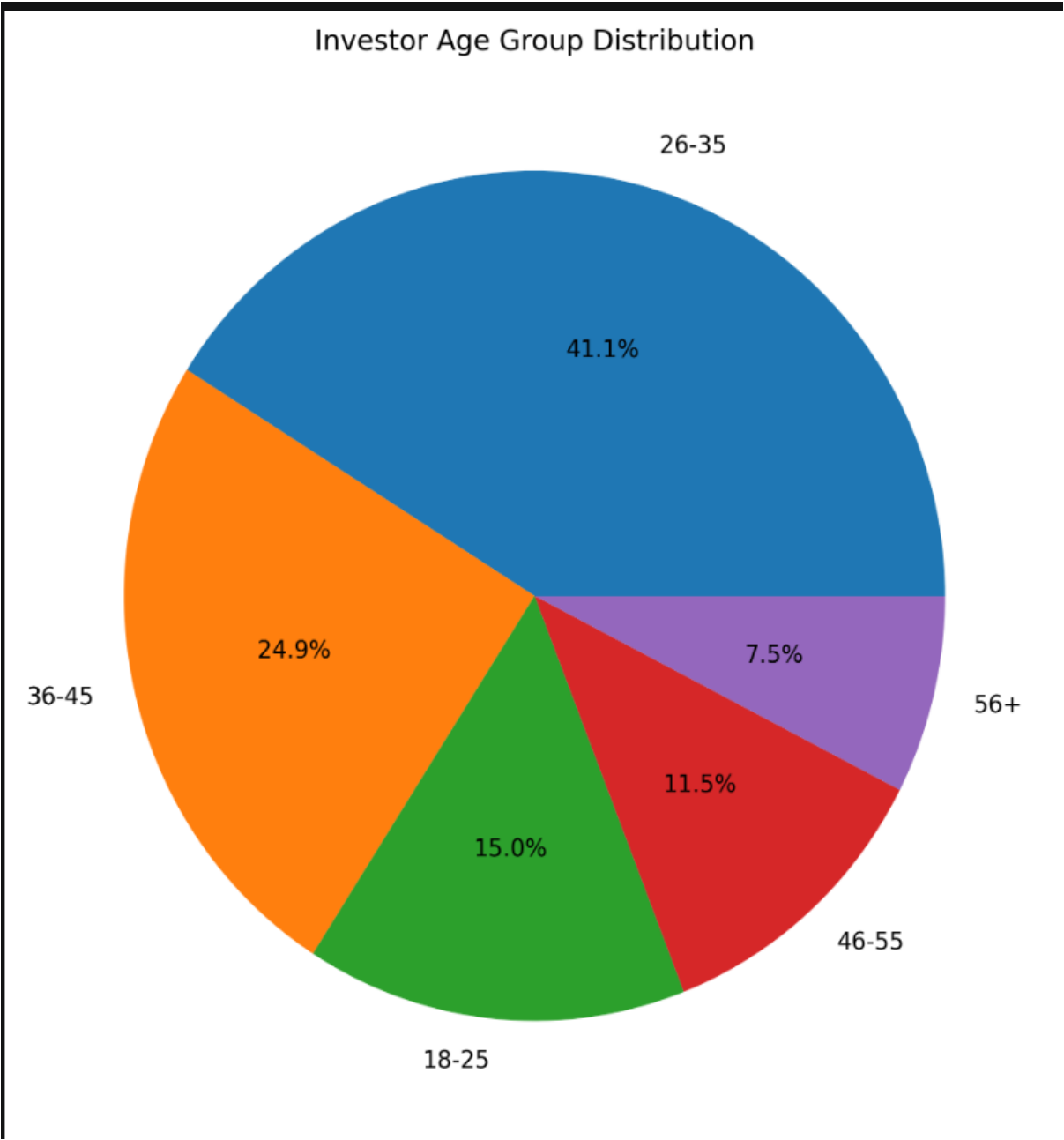


Figure 3: Investor Age Group Distribution



7. Fund Performance Analytics

Fund performance analytics focused on evaluating risk-adjusted returns, benchmark performance, and long-term investment growth. Multiple quantitative metrics were calculated to compare mutual fund schemes across different categories.

Metrics Calculated:

Compound Annual Growth Rate (CAGR):

CAGR was calculated for 1-year, 3-year, and 5-year periods to evaluate annualized growth rates.

Sharpe Ratio:

The Sharpe Ratio was used to measure excess return per unit of risk. Higher Sharpe values indicate superior risk-adjusted performance.

Sortino Ratio:

Sortino Ratio was used to evaluate downside-risk-adjusted returns by considering only negative return volatility.

Alpha:

Alpha measures a fund's excess return relative to its benchmark after adjusting for market risk.

Beta:

Beta quantifies sensitivity of fund returns to market movements. Values above 1 indicate higher volatility than the benchmark.

Maximum Drawdown:

Maximum Drawdown was used to measure the largest decline from peak portfolio value, providing insights into downside risk.

Benchmark Comparison:

Top-performing funds were compared against NIFTY 50 and NIFTY 100 benchmarks to evaluate consistency and relative performance.

The analysis identified high-performing funds that consistently delivered strong risk-adjusted returns while maintaining manageable levels of volatility and drawdown risk.

Figure 4: Benchmark Comparison

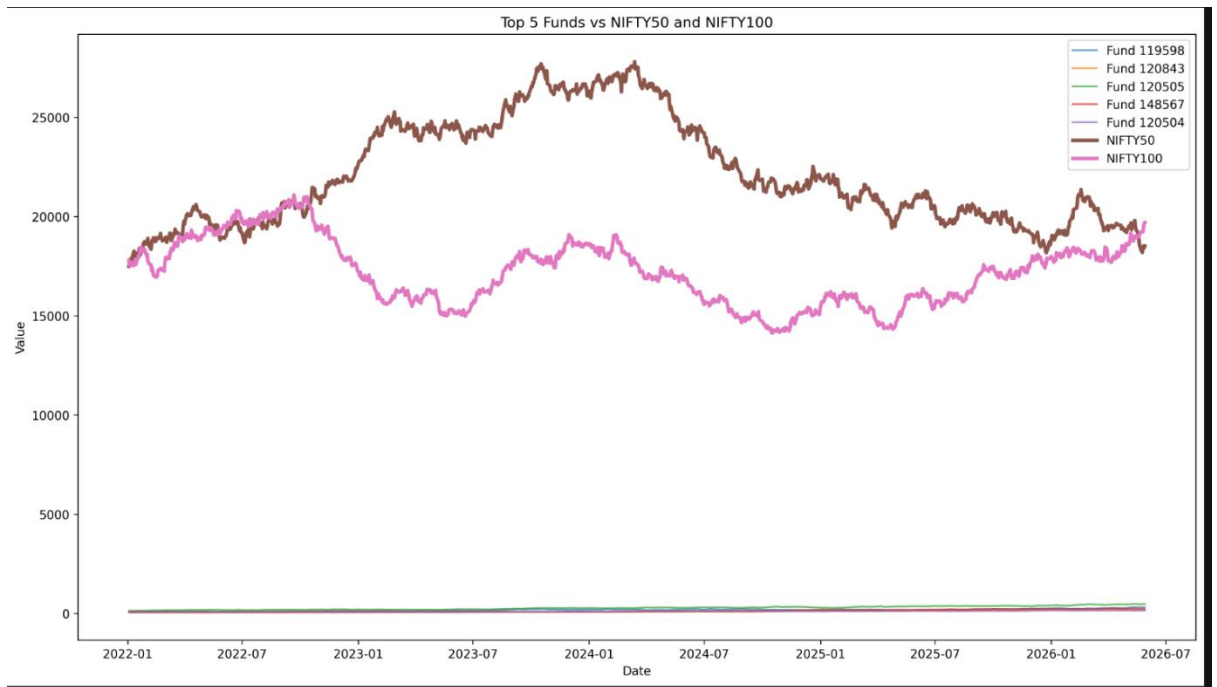


Figure 5: Sharpe Ratio Ranking

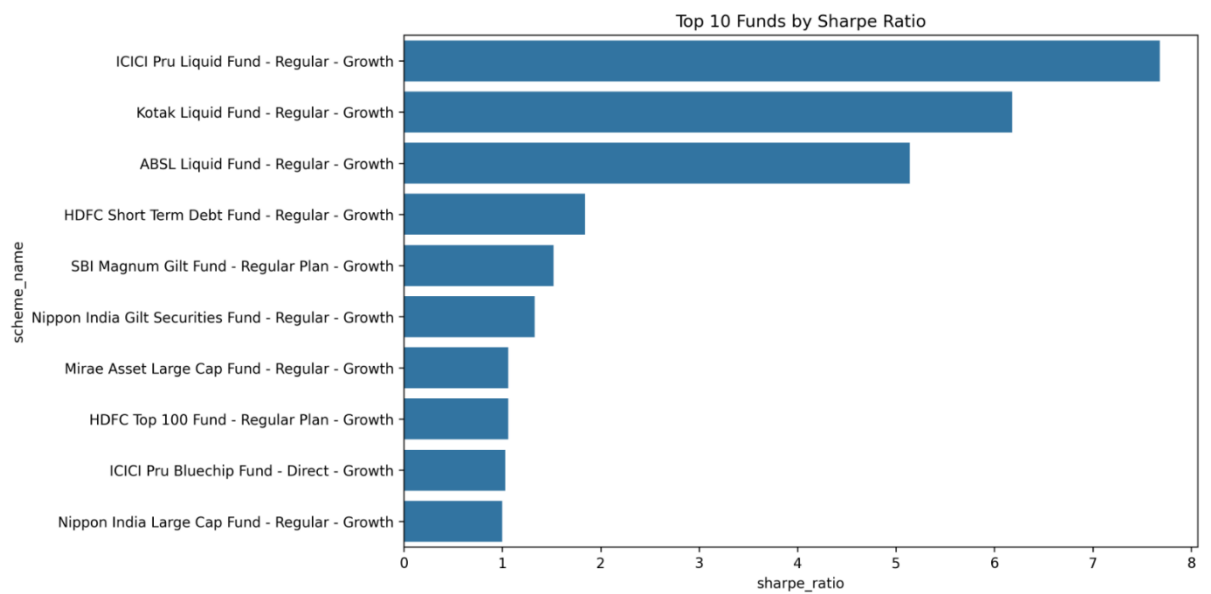
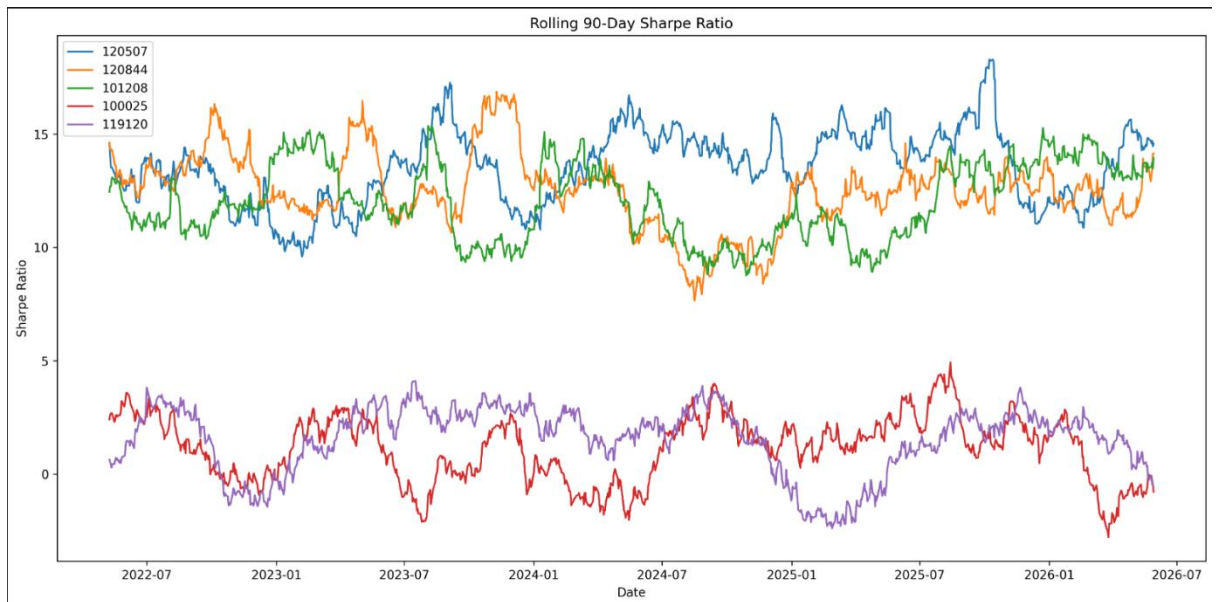


Figure 6: Rolling Sharpe Ratio



8. Advanced Risk Analytics

Advanced risk analytics were performed to assess downside risk, portfolio concentration, and performance stability.

Value at Risk (VaR):

Historical 95% VaR was calculated to estimate potential portfolio losses during adverse market conditions.

Conditional Value at Risk (CVaR):

CVaR was used to measure expected losses beyond the VaR threshold, providing a more conservative assessment of downside risk.

Rolling Sharpe Ratio:

A 90-day rolling Sharpe Ratio was calculated to analyze changes in risk-adjusted performance over time.

Investor Cohort Analysis:

Investors were grouped based on their first transaction year to analyze investment behavior, contribution levels, and fund preferences.

SIP Continuity Analysis:

Investor contribution frequency was evaluated to identify investors at risk of discontinuing SIP investments.

Herfindahl-Hirschman Index (HHI):

HHI was used to measure portfolio concentration by analyzing stock-level allocation weights within mutual fund portfolios.

Fund Recommendation Engine:
A rule-based recommendation engine was developed to suggest suitable mutual funds based on investor risk appetite and historical performance metrics.

These advanced analytics provided deeper insights into fund risk, portfolio diversification, and investor engagement patterns.

Figure 7: Maximum Drawdown Analysis

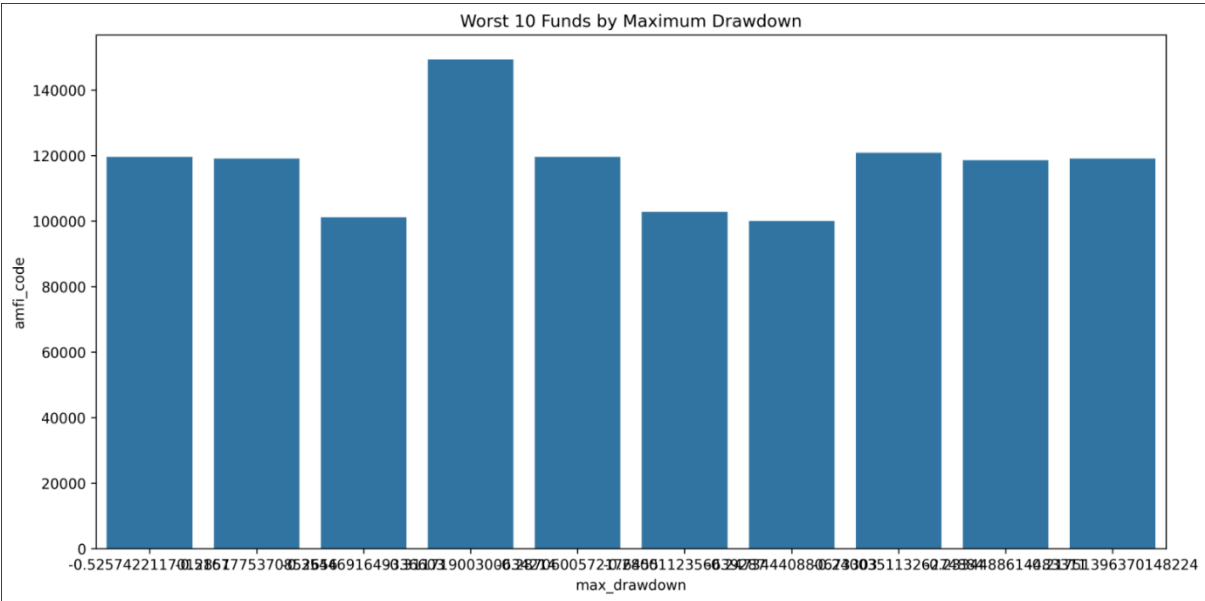
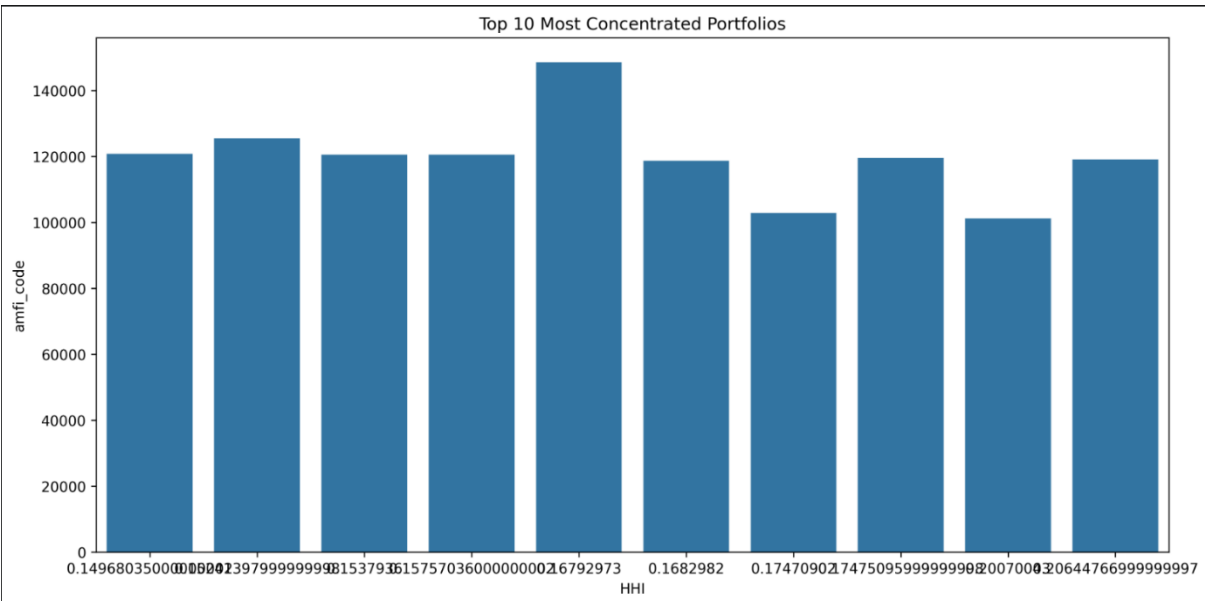


Figure 8: Portfolio Concentration (HHI)



9. Dashboard Development

The final business intelligence solution was developed using Microsoft Power BI. The dashboard was designed to provide an interactive and user-friendly interface for analyzing mutual fund performance, investor behavior, industry growth, and SIP trends.

Dashboard Features:

- Industry Overview page displaying industry AUM, SIP inflows, folio counts, and fund house analysis.
- Fund Performance page featuring risk-return analysis, benchmark comparison, fund scorecards, and category-wise filtering.
- Investor Analytics page presenting transaction trends, demographic insights, geographic distribution, and investment behavior patterns.
- SIP and Market Trends page analyzing SIP inflows, category inflows, benchmark movements, and market trends.

Interactive slicers were implemented to allow filtering by fund house, category, plan type, investor demographics, and geographic attributes. The dashboard enables users to quickly identify trends, compare fund performance, and evaluate investment opportunities through dynamic visualizations.

Figure 9: Industry Overview Dashboard

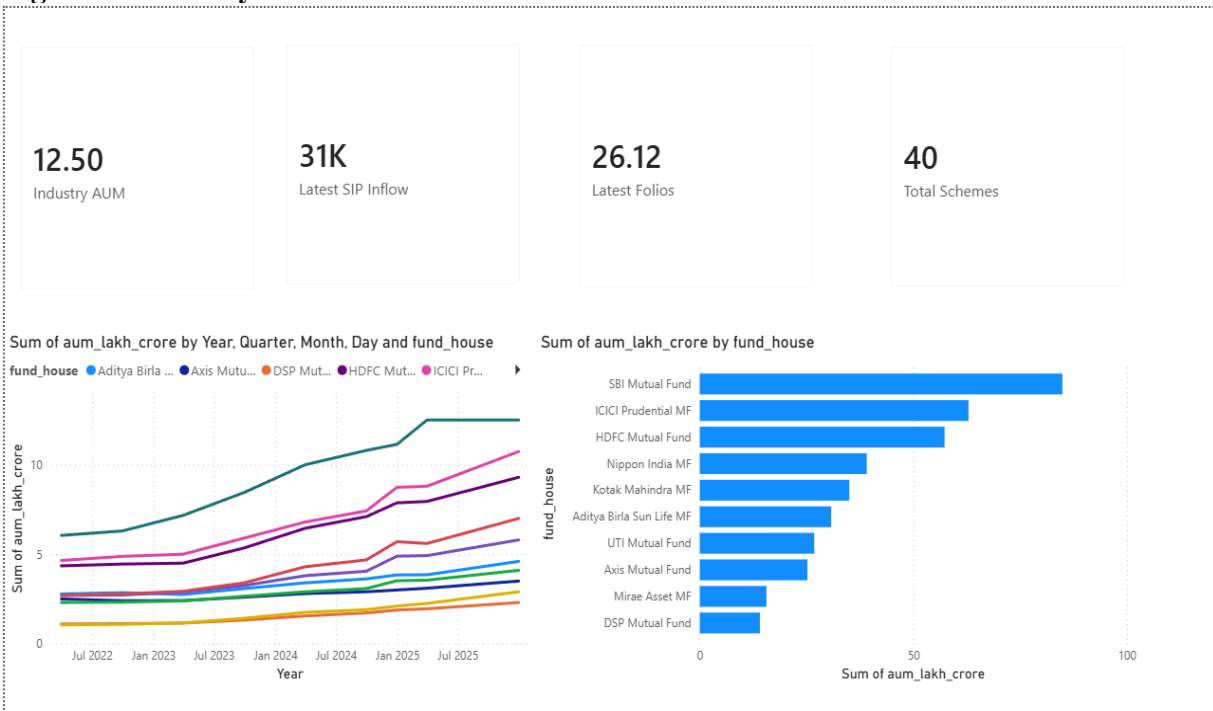


Figure 10: Fund Performance Dashboard

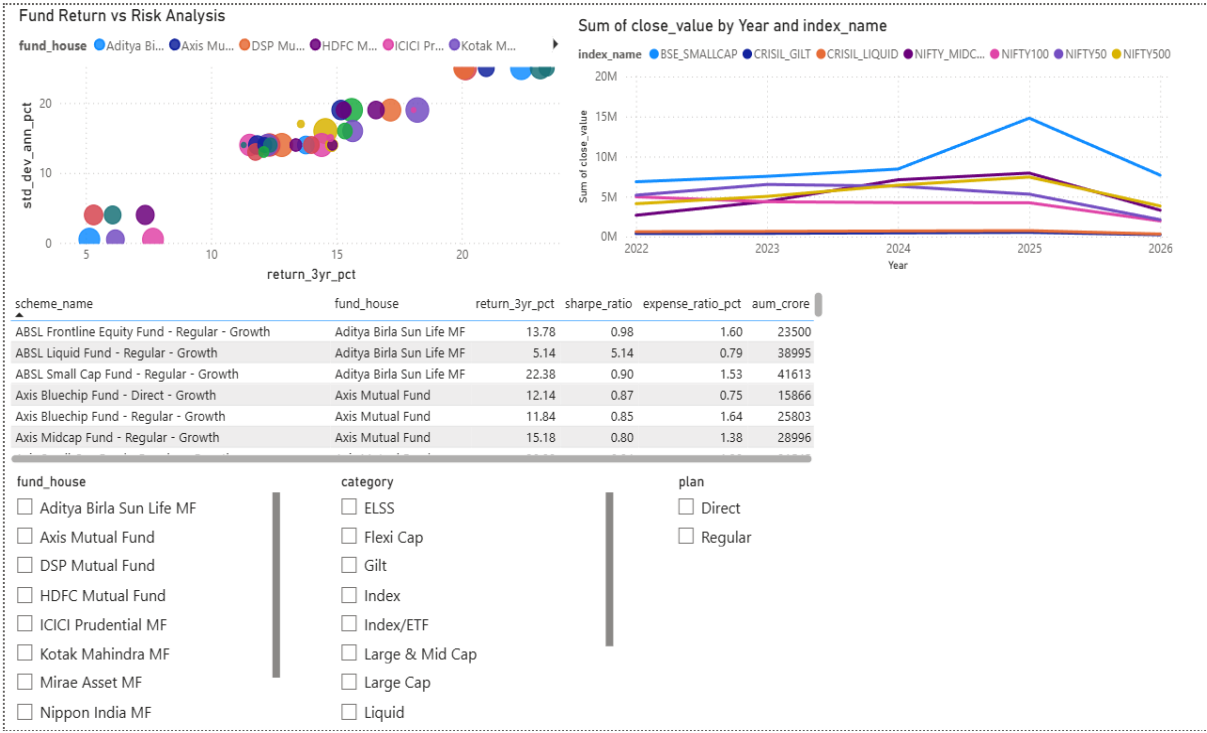


Figure 11: Investor Analytics Dashboard

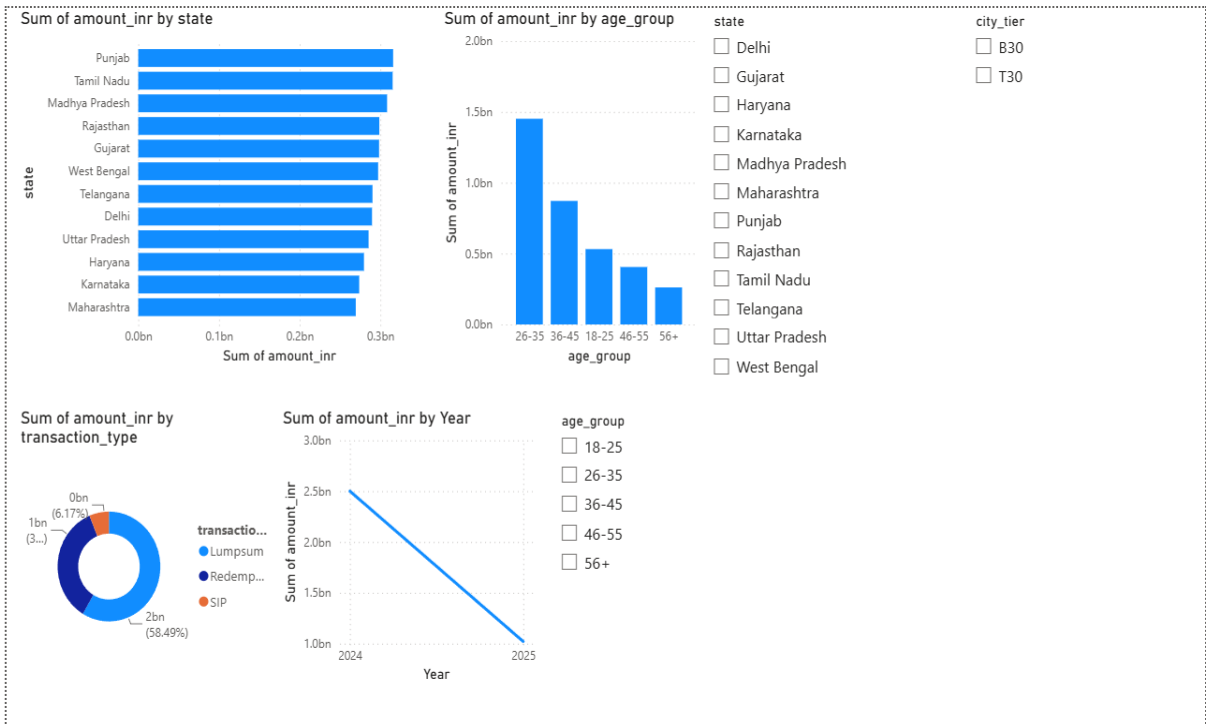
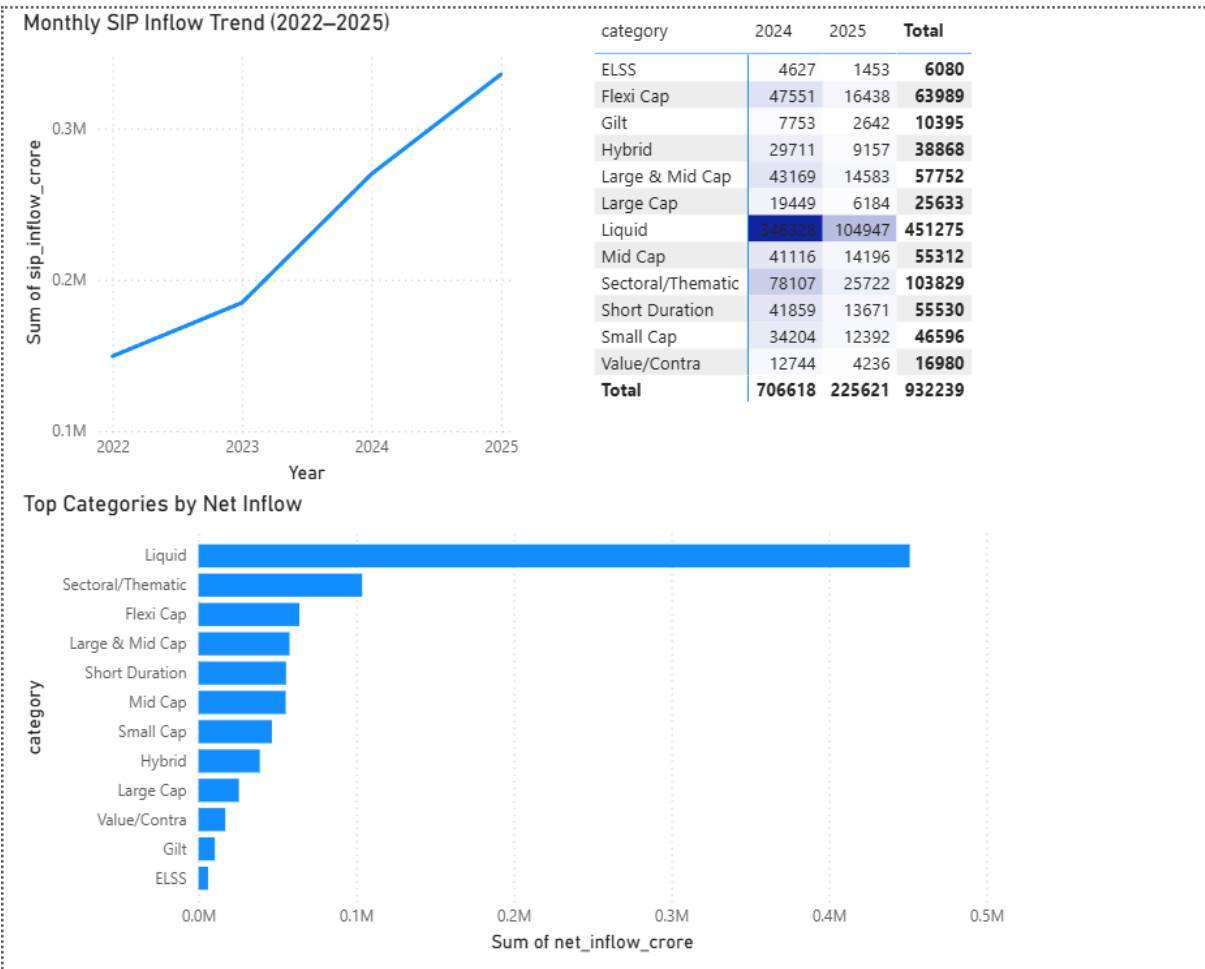


Figure 12: SIP and Market Trends Dashboard



10. Key Findings

The analysis generated several important insights regarding mutual fund performance, investor behavior, and industry trends.

Key Findings:

1. Industry AUM experienced significant growth between 2022 and 2025, reflecting increased investor participation in mutual funds.
2. SIP inflows consistently increased throughout the study period and reached record levels by 2025, demonstrating growing adoption of systematic investing.
3. Several equity-oriented schemes delivered superior risk-adjusted returns as measured by Sharpe Ratio and Sortino Ratio.
4. Rolling Sharpe Ratio analysis indicated that top-performing funds maintained relatively stable risk-adjusted performance over time.
5. Historical VaR and CVaR analysis highlighted meaningful differences in downside risk across mutual fund schemes.
6. Investor demographic analysis revealed strong participation from younger investors, particularly within the 26–35 age group.

7. SIP Continuity Analysis identified a significant proportion of investors with irregular contribution patterns, indicating potential risk of SIP discontinuation.
8. HHI analysis showed varying levels of portfolio concentration, with some funds exhibiting greater diversification than others.
9. Benchmark comparison demonstrated strong relationships between fund performance and major market indices such as NIFTY 50 and NIFTY 100.
10. The recommendation engine successfully identified suitable mutual funds based on investor risk profiles and historical risk-adjusted returns.

11. Limitations

Although the project successfully achieved its objectives, several limitations should be considered.

- The analysis is dependent on the quality and completeness of available datasets.
- Some calculations rely on historical performance and may not accurately predict future returns.
- The recommendation engine uses rule-based logic and does not incorporate machine learning techniques.
- Portfolio concentration analysis is limited to available holdings data and may not reflect real-time portfolio changes.
- Market conditions, regulatory changes, and macroeconomic events were not explicitly modeled within the analysis framework.
- Dashboard insights are based on historical data and should be interpreted alongside broader market research and investment considerations.

12. Recommendations

Based on the analysis performed throughout the project, the following recommendations are proposed:

- Expand the recommendation system using machine learning algorithms and personalized investor profiling.
- Integrate real-time market data feeds for continuous monitoring and analysis.
- Enhance investor risk assessment models using additional behavioral and financial indicators.
- Develop predictive analytics capabilities for forecasting fund performance and investor churn.

- Introduce automated alert systems for SIP discontinuation risk and unusual portfolio concentration.
- Expand benchmark coverage to include additional market indices and international funds.
- Deploy the analytics platform as a cloud-based business intelligence solution to improve accessibility and scalability.

13. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrates the practical application of data analytics, financial modeling, and business intelligence techniques within the mutual fund industry.

The project integrated data engineering, exploratory analysis, performance measurement, risk analytics, investor behavior analysis, dashboard development, and recommendation systems into a unified analytical framework. Through the use of Python, SQL, SQLite, and Power BI, meaningful insights were generated from complex financial datasets.

The resulting solution provides a comprehensive view of mutual fund performance, industry growth, investor participation, and portfolio risk. The project not only fulfills the defined objectives but also establishes a strong foundation for future enhancements involving machine learning, predictive analytics, and real-time financial intelligence systems.

14. References

References

1. Association of Mutual Funds in India (AMFI)
2. National Stock Exchange of India (NSE)
3. Securities and Exchange Board of India (SEBI)
4. Microsoft Power BI Documentation
5. Python Documentation
6. Pandas Documentation
7. NumPy Documentation
8. Matplotlib Documentation
9. Seaborn Documentation

10. SQLite Documentation